



ADVANCED PREDICTIVE ANALYTICS

10 Steps To Build An Advanced
Predictive Model

1. Building a Predictive Analytics Team

What to Do: Assemble a multidisciplinary team that includes data scientists, analysts, domain experts, IT professionals, and business leaders.

Why It's Important: Each team member brings unique expertise—data scientists develop the models, domain experts ensure relevance, and IT supports infrastructure.

Example: A bank forming a team for credit risk modeling may include a data scientist, a loan officer, and an IT specialist to ensure seamless integration.



Which are the right questions?

Which aspects of your business do you want predictive analytics to improve



Which is the right data to include in the analysis?

Should your focus be on the efficiency of your business processes? The demographics of your customers? Which body of data stands out as the most critical?)



Who are the business stakeholders ?

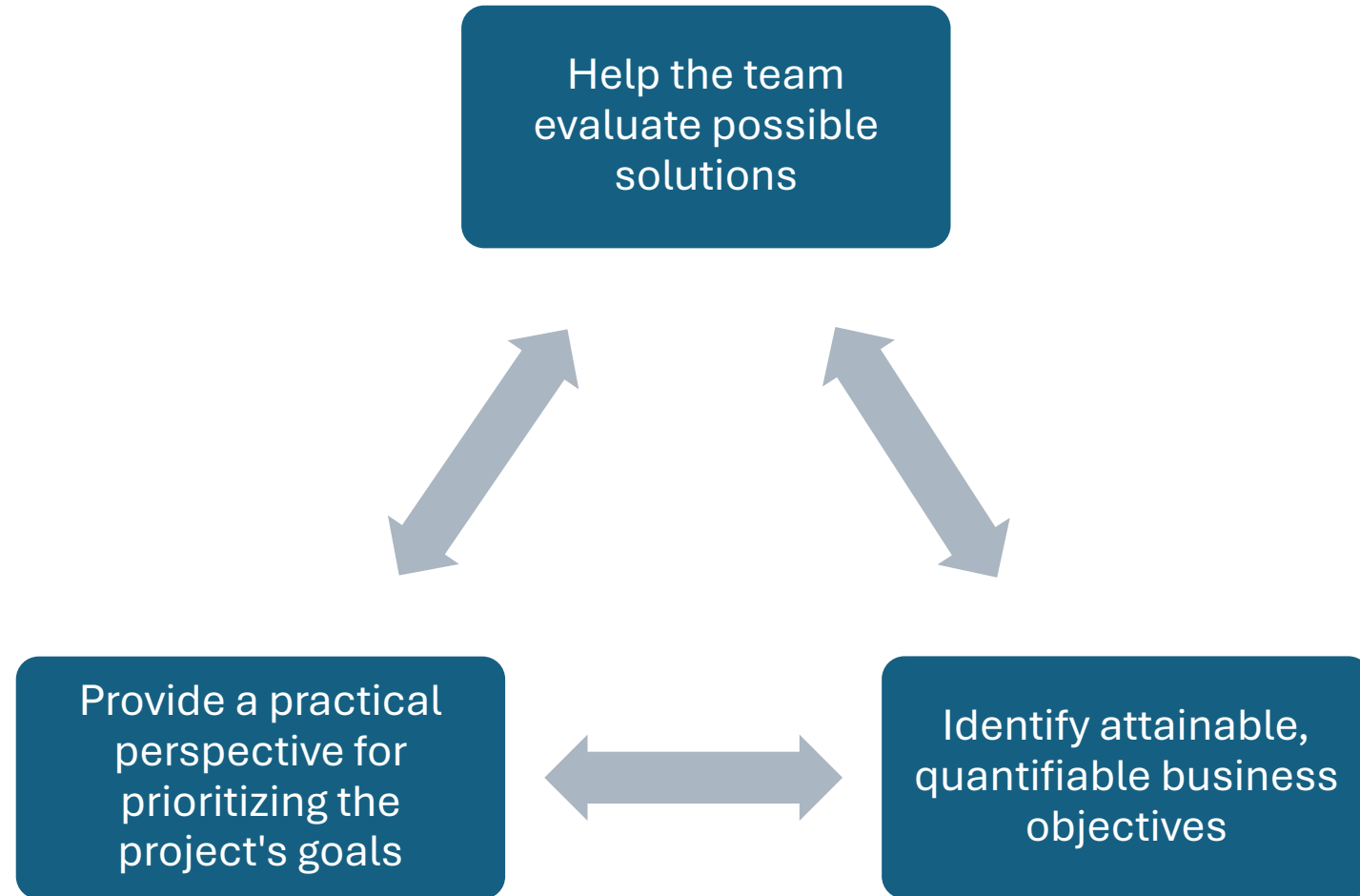
How can they benefit from the insights gained from your predictive analytics project

2. Setting the Business Objectives

What to Do: Clearly define the goal the predictive model will achieve with the help of stakeholder's domain knowledge and their expectations on the outcome of project. This could involve solving a problem, improving efficiency, or seizing a market opportunity.

Why It's Important: Without clear objectives, models may lack focus and fail to deliver actionable insights.

Example: An insurance company might aim to reduce claims fraud by 15% through early detection using predictive analytics.



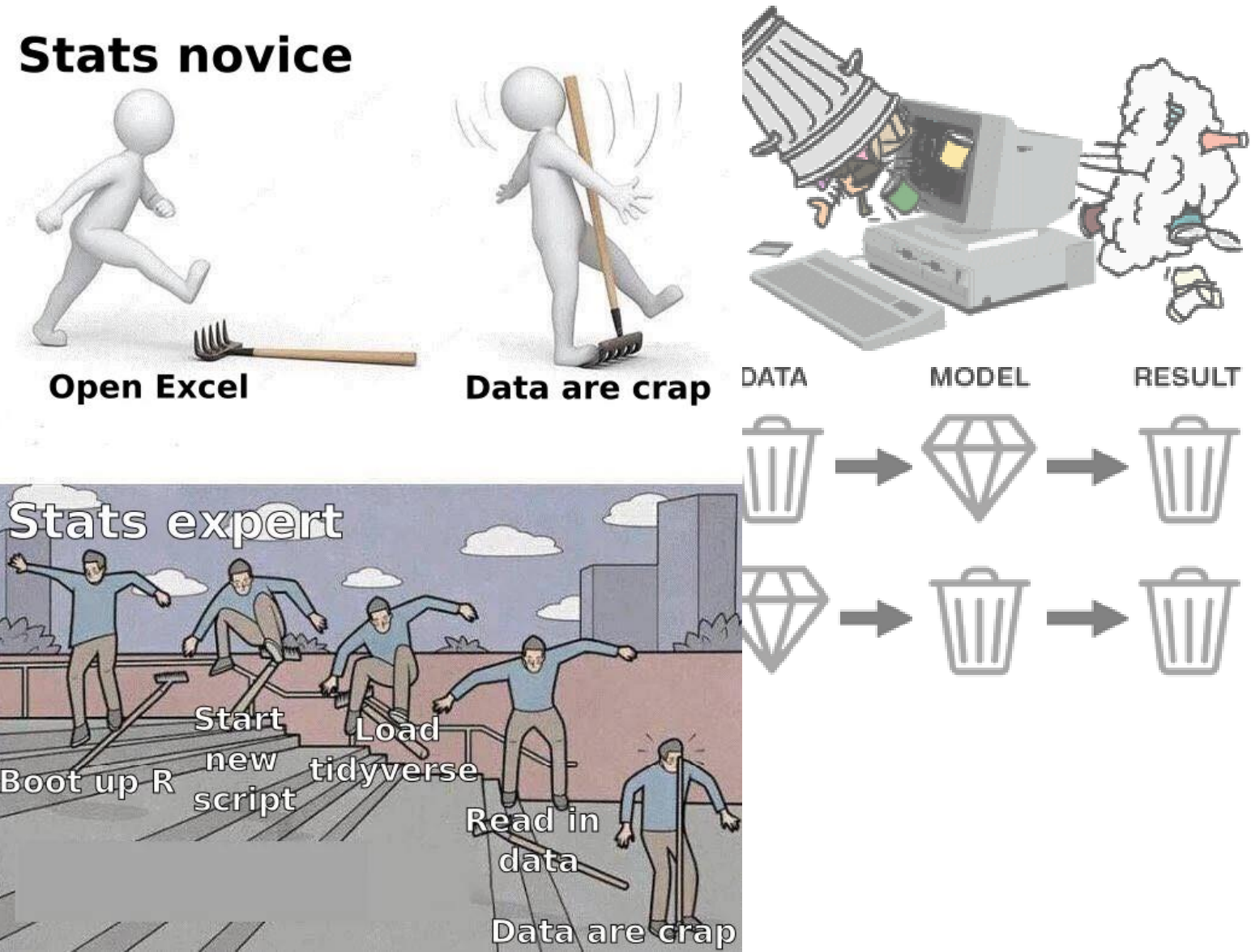
3.Preparing Your Data

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4. Avoiding “Garbage In, Garbage Out”

- What to Do:** Ensure the quality of data used for model development. Verify that data is accurate and consistent with the problem.
- Why It’s Important:** Poor data quality leads to poor predictions, which can mislead decision makers.
- Example:** For predicting loan default, the accuracy of income data, credit score, and employment history is crucial.



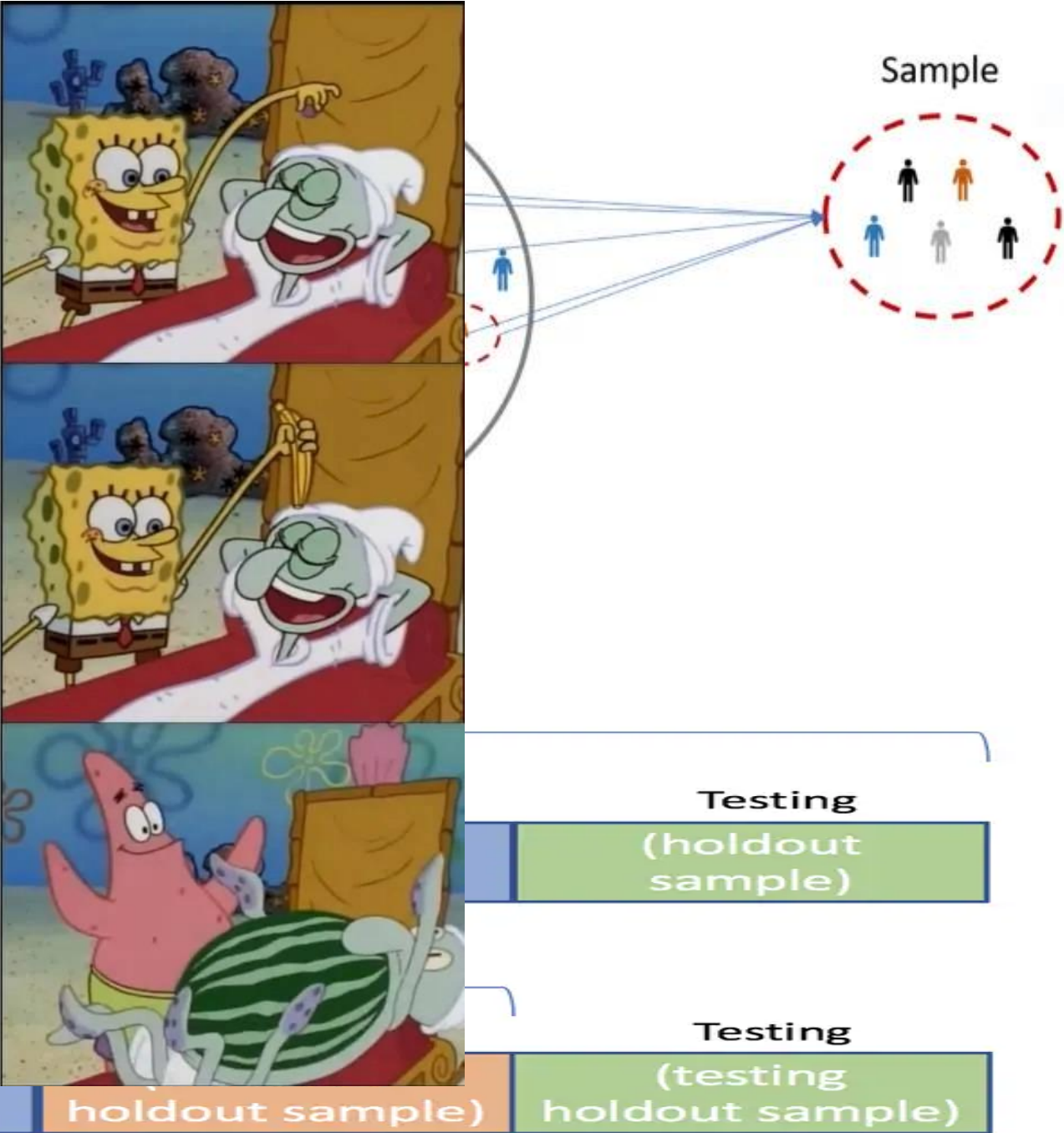
5. Sampling Your Data

- What to Do:** Select a representative sample of the data for training and testing.
- Why It's Important:** Sampling is important because it reduces computational requirements and ensures the model performs well on new datasets.
- Example:** A credit card company might sample 10% of its transactions for fraud detection model training.

Training Set

Validation Set

Test Set



6. Creating Quick Victories

- What to Do:** Start with **Incremental** demonstrate the poten
- Why It's Important:** Q buy-in from stakeholde projects.
- Example:** A bank migh optimize ATM cash repl complex customer life



to building the model — model, fine-tuning it lts, and then trying the ill allow you to evaluate orithms used in your ose best suited to your g your model iteratively ome decisions and

Iterative



Evaluation

Testing

Deployment

7. Fostering Change

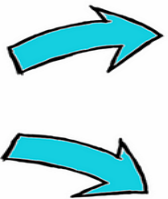
- **Shift Leadership**
 - Educate leadership
 - Demonstrate
- **Promote Organizational Change**
 - Publicize success
 - Highlight the benefits of the new organization without it.
- **Foster Cultural Change**
 - Integrate change into business processes
 - Recruit team members who communicate the change
- **Overcome Resistance**
 - Address sources of resistance
 - Build trust and credibility through recommendations
- **Educate and Train**
 - Invest time in training
 - Focus on core values.



8. Evaluating Your Model

- What to Do:** Test the model using unseen data to evaluate its accuracy, precision, recall, and other performance metrics.
- Why It's Important:** Evaluation ensures the model performs reliably and accurately.
- Example:** A credit scoring model's ability to correctly predict loan defaults.

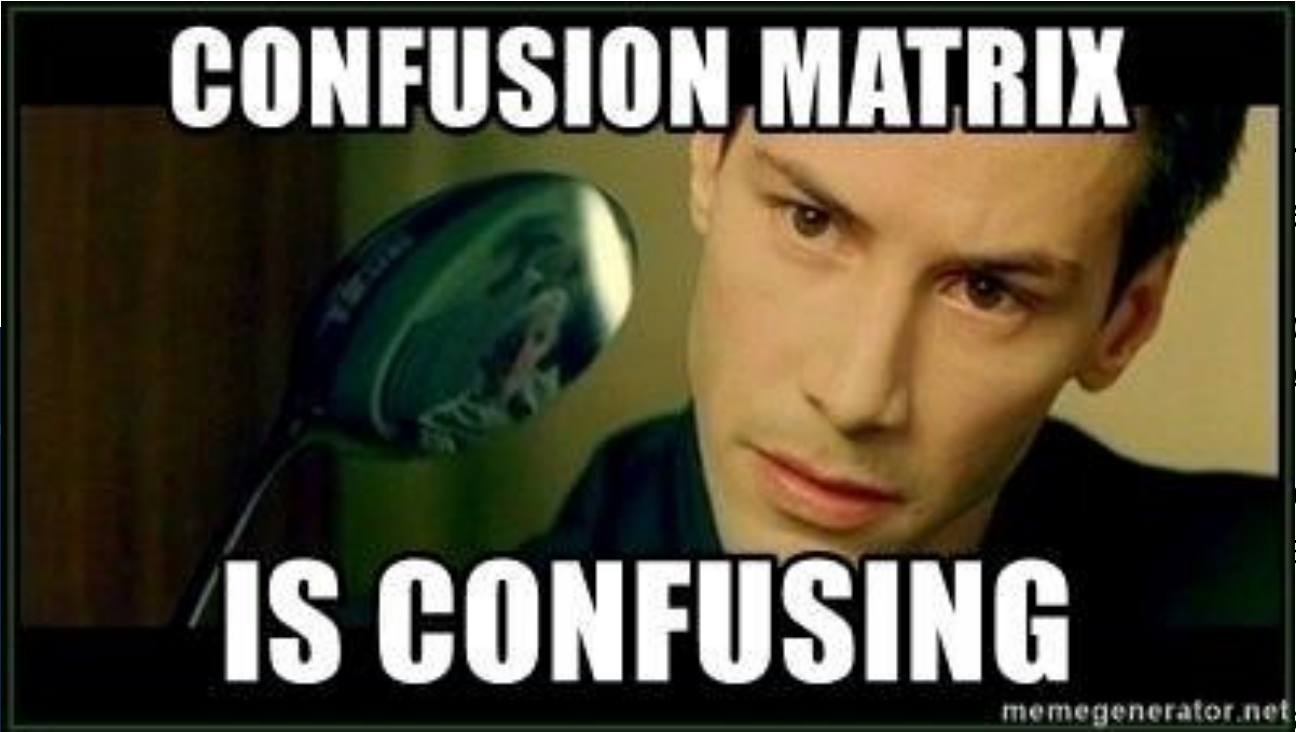
MODEL



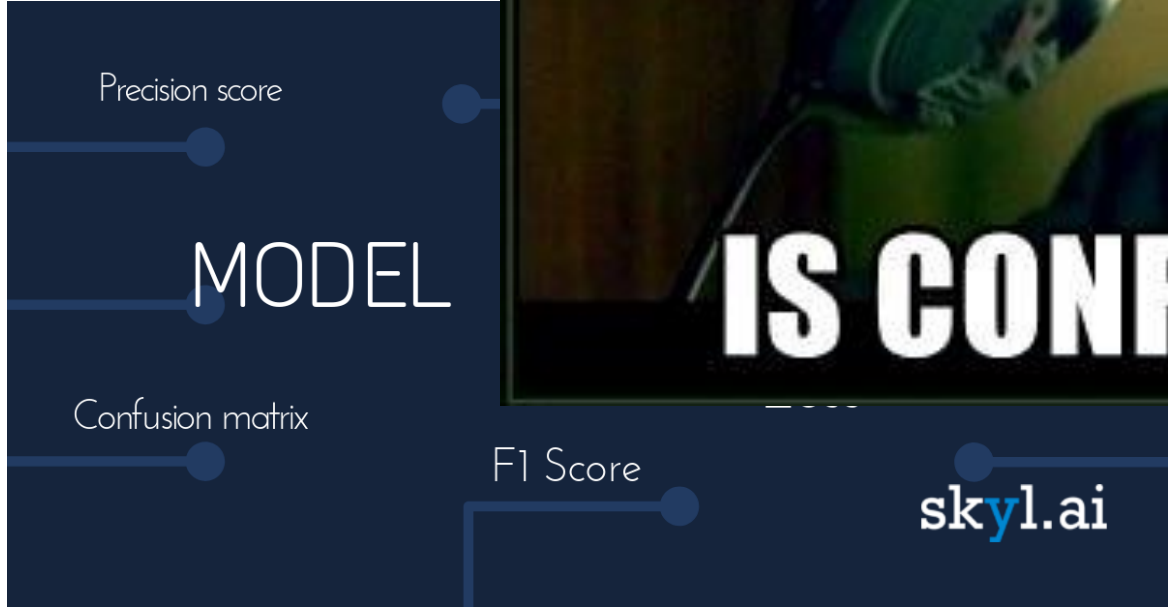
GOOD?



BAD?

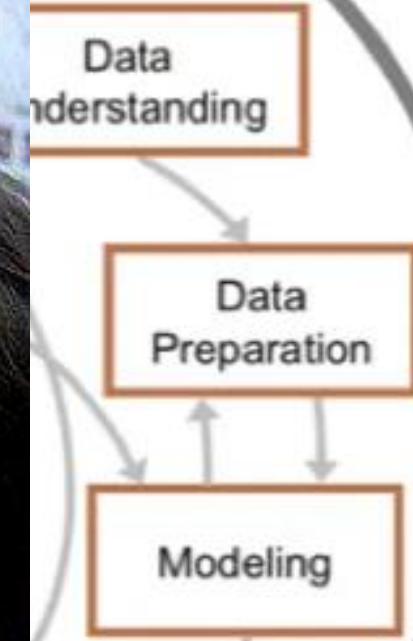


From these two distinct angles:
A business analyst should evaluate the model's performance and the accuracy of its predictions. Are the insights derived from the model easier for you to make decisions based on? Are you spending more time or less time because of these new insights? Are the algorithms used and the methods applied. Are the algorithms chosen optimal for the model's purpose? Are the insights being generated fast enough to produce actionable advantages?



9. Building Deployable Models

- **What to Do:** Develop models that can be easily deployed to production systems and work with various data sources, algorithms and environments, and deployment platforms.
- **Why It's Important:** Models need to be operationalized to be useful.
- **Example:** A bank's credit risk model, loan approval model, customer relationship management system, or real-time recommendation system.

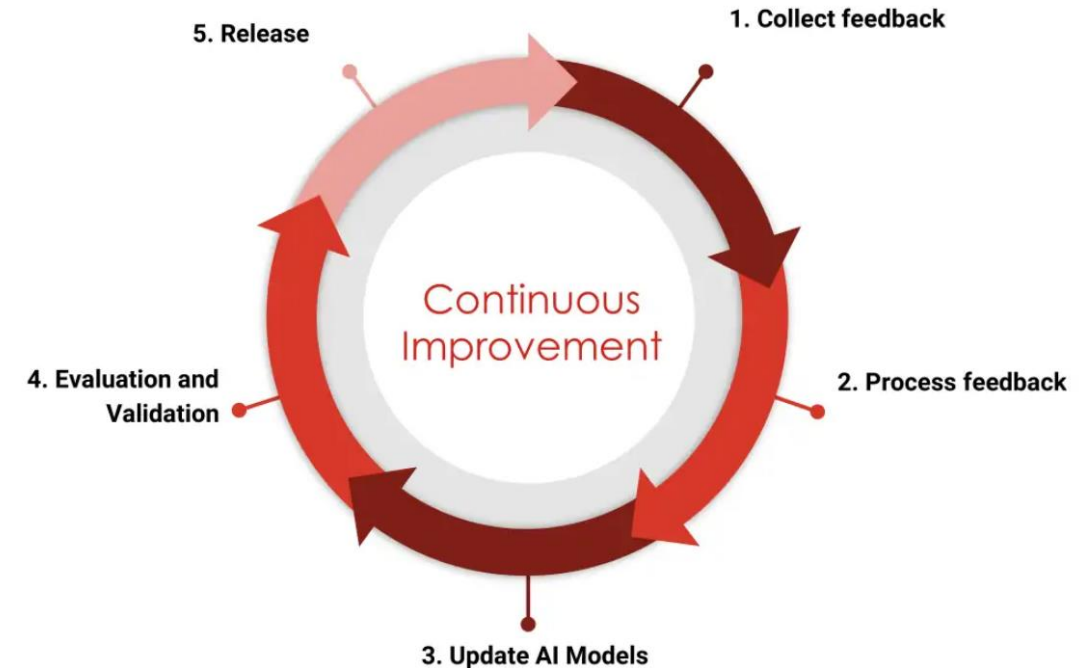


10. Updating Your Model

- What to Do:** Continuously monitor and retrain the model using fresh data to ensure it remains accurate and relevant as conditions change.
- Why It's Important:** Data patterns evolve over time, and outdated models can produce inaccurate predictions.
- Example:** A fraud detection model in a bank might require frequent updates to account for new fraud tactics.

Suggestions :

- Stay on top of changing conditions by retraining and testing the model regularly; enhance it whenever necessary.
- Monitor your model's accuracy to catch any degradation in its performance over time.
- Automate the monitoring of your model by developing customized applications that report and track the model's performance





THANK YOU!!